INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

COURSE TITLE: BACHELOR OF TECHNOLOGY

SUBJECT: PYTHON PROGRAMMING LABORATORY

PROJECT TITLE: STUDENT PERFORMANCE AND STUDY-HOURS DASHBOARD

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Abstract

This project develops a Python-based desktop dashboard to examine whether reported weekly study hours are associated with student assessment scores. It uses anonymised student identifiers, validates scores and study hours, converts scores to letter grades, supports CSV import and export, summarises performance by course, and displays statistical charts. The implementation combines object-oriented programming and standard CSV handling with NumPy, Pandas, Matplotlib and Tkinter.

The included demonstration dataset is synthetically generated, not collected from actual students. From 125 input rows, 123 pass validation and two are rejected. Using a 1.5 × IQR rule on either study hours or scores, four observations are flagged as outliers. Pearson's correlation is 0.328 using all valid records and 0.609 after excluding flagged records. These results indicate a positive association in this simulated dataset, while also demonstrating the sensitivity of the linear correlation to unusual observations. The project does not establish a causal effect of studying on marks.

Keywords

Student performance; study hours; Python; CSV; Tkinter; NumPy; Pandas; Matplotlib; correlation; outlier analysis.

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1. Introduction

Student records frequently contain assessment marks and estimates of time spent studying. When these records are stored in a spreadsheet, comparing distributions, courses and unusual observations can be time-consuming. A small analytical dashboard can standardise data validation and present useful summaries for exploration.

The developed application is a Python desktop interface. It displays key performance indicators, a study-hours-versus-score scatter plot, score and grade distributions, course comparisons, an outlier table, and a written statistical interpretation. Users can work with the preloaded synthetic dataset or import an anonymised CSV file.

2. Problem Statement and Objectives

Problem statement: Investigate assessment scores and reported study hours using anonymised records. Build score/grade functions, handle records in dictionaries and CSV, implement a StudentRecord class, compute array statistics with NumPy, aggregate by course with Pandas, and draw scatter plots and histograms. Evaluate whether the apparent relationship persists after handling outliers, without interpreting correlation as causation.

The principal objectives are:

- Validate input records and convert numerical marks into grades A, B, C, D or F.
- Read and export structured student data in CSV form.
- Apply a StudentRecord class to encapsulate record fields, validation and grade conversion.
- Calculate measures of central tendency and dispersion and compare course-level performance.
- Display interactive charts and identify unusually high or low measurements.
- Compare Pearson and Spearman associations before and after excluding flagged outliers.
- Provide an understandable written interpretation and note methodological limitations.

3. Software and System Requirements

Component	Purpose
Python 3	Coreprogram execution
Tkinter / ttk	Desktop graphical user interface
csv, random,math	CSV handling, sample generation and validation
NumPy	Arrays, percentiles, numerical analysis and trend estimation
Pandas	Tabular records, filtering, group-by summaries and correlations
Matplotlib	Scatter plots, histograms and bar charts
Python IDLE	Runthe desktopprogram using RunModule(F5)

The programis suitable for a desktop system with a working Python installation and therequiredthird-party libraries. Tkinter is normally supplied with standard Python installations, subject to platform packaging.

4. System Design and Methodology

4.1 Input data and schema

Each record contains four fields: student_id (an anonymised identifier), course (subject or programme label), study_hours (reported study hours per week), and score (assessment marks on a 0–100 scale). The program derives grade from score rather than requiring it as a CSV input column.

4.2 Data preparation

On startup, the Tkinter version creates a reproducible example dataset using a fixed random seed. It attempts to validate 125 rows, including deliberately problematic examples. Alternatively, Open CSV allows a user to

choose a CSV dataset with the required column names. The program checks that identifiers and course names are nonempty, study hours lie between 0 and 168, and scores lie between 0 and 100. Invalid rows are skipped and reported.

4.3 Analysis workflow

1. Read or generate student records.
2. Validate each row using StudentRecord and convert records to dictionaries.
3. Create a Pandas DataFrame and apply the selected course filter.
4. Compute descriptive statistics and assign letter grades.
5. Flag outliers using the interquartile-range (IQR) rule.
6. Calculate Pearson and Spearman correlations for the full and non-outlier groups.
7. Refresh charts, record table, summary indicators and written interpretation.

4.4 Statistical measures

The mean and median describe the centre of the score distribution; the sample standard deviation describes spread. The interquartile range is IQR = Q3 − Q1. A value is flagged when it falls below Q1 − k(IQR) or above Q3 + k(IQR), where k defaults to 1.5. A row is flagged if its study hours or score is outside the corresponding interval.

Pearson's r measures linear association between study hours and score. Spearman's ρ measures association between their ranks. Both range from −1 to +1 when defined. The program compares each measure before and after flagged observations are excluded; these comparisons are descriptive rather than proofs of causal influence.

5. Implementation of Major Modules

Module / function	Functionality
is_valid_score(), score_to_grade()	Scorevalidation and grade mapping
StudentRecord, from_dict(), to_dict()	Object-oriented records, conversion and field validation
generate_sample_data(), validate_rows(), load_csv()	Synthetic data, input validation and CSV import
iqr_mask(), analysis_for(), correlation()	Outlier detection and correlation comparison
Dashboard._build(), refresh()	GUI layout, filters, key indicators and updated views
draw_scatter(), draw_scores(), draw_courses()	Embedded Matplotlib visualisations
interpretation()	Automated descriptive discussion and non-causation warning
open_csv(), save_csv()	Load user-supplied records and export current filtered data

5.1 Interface layout

The main window includes an Open CSV button, a Use sample data button, a course selector, an IQR threshold selector and a checkbox for hiding outliers in the charts. Five notebook tabs provide: Study hours vs scores, Scores & grades, Course comparison, Outliers & records, and Written analysis. Five metric cards display the student count, mean score, median score, mean study hours and outlier count.

5.2 Grade mapping

Grade thresholds: A = 70–100; B = 60 to below 70; C = 50 to below 60; D = 40 to below 50; F = below 40. The stated intervals describe the effective decision thresholds; fractional scores are allowed. Values outside 0–100 are rejected.

6. Results and Discussion

6.1 Demonstration dataset and descriptive findings

The following results were computed from the fixed-seedsynthetic dataset included in the submitted Tkinter program.They will differ if a separate CSV is imported,afilter is applied, or the IQR threshold is changed.

Indicator	Observed value
Input rows	125
Valid records	123
Rejected rows	2
Average assessment score	60.69 / 100
Median assessment score	61.60 / 100
Sample standard deviation of score	14.63
Mean reported study hours	12.80 hours / week
Q1 / Q3 of score	51.90 / 71.50 19.60
Score IQR	4 of 123 (3.3%)
Outliers flagged (k = 1.5)	

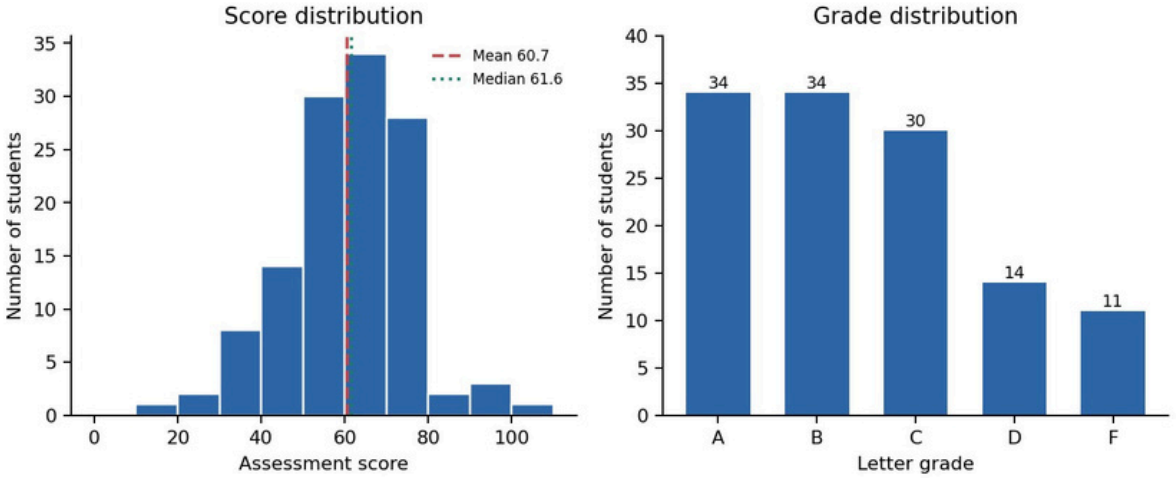


Figure 1. Score histogram and grade counts for the generated sample.

6.2 Course-wise summaries

Course	Students	Meanscore	Meanhours/week
Biology	35	62.47	12.91
Business	21	61.13	11.35
Computing	40	57.53	11.50
Maths	27	62.71	15.69

In thisdemonstration sample,Mathematics has the highestaverage score (62.71), whileComputing has the lowest(57.53). These groupdifferences are descriptive andshould not be interpreted asintrinsic differences in difficulty or teaching quality.

6.3 Relationship between study hours and scores

Correlation statistic	All valid records	Without flagged outliers
Pearson r	0.328	0.609
Spearman ρ	0.608	0.670

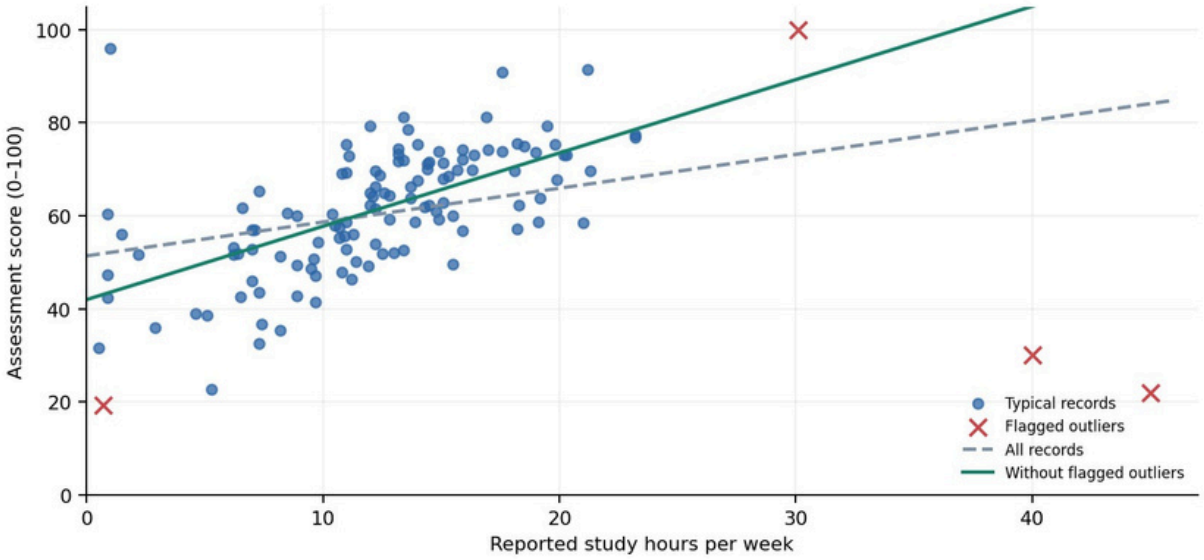


Figure 2. Study hours and scores, showing flagged observations and linear fits.

Pearson's r increases from 0.328 to 0.609 after removing the four IQR-flagged records,an absolute change of about 0.281. Spearman's rank correlation is comparatively more stable (0.608 versus0.670). The positive association remains in the filtered sample but is substantially affected in magnitude bythe flagged cases. Accordingly, the original linear correlation should be reported together with the outlier-sensitivity check.

6.4 Outliers and interpretation

Student ID	Course	Hours/week	Score
S065	Computing	30.1	100.0
S088	Computing	0.7	19.3
S901	Maths	45.0	22.0
S902	Biology	40.0	30.0

TheIQR rule identifies statisticalextremes, not necessarilymistakes. For example, anunusually high study-hour report may be a genuineobservation. Removingsuch records is a sensitivityanalysis and should notautomatically be treated asadata-cleaning requirement.

Theproject does not show thatlonger study time causes higher marks. Prior ability, attendance, course content, learning methods, motivation and measurement error may contribute to the observed association. Moreover, the built-in recordsaregenerated by a mathematical procedure, so the numerical results are examples of dashboard behaviourrather than empirical evidence about students.

7. Testing and Validation

Test case	Expected behaviour	Observed on sample
Sample startup	Build records andpopulate dashboard	123accepted;2rejected Rejected
Invalid score (140)	Reject score outside 0–100	Rejected
Blank study hours	Reject nonnumeric study hours	4 flagged
Default IQR (1.5)	Flag extreme scores/hours	Implemented in GUI refresh
Course selector CSV	Recalculate for selected course	Implemented using DictReader
import	Read required columns and report bad rows	Implemented as chart filter
Hide outliers	Replot excluding flagged records	Implemented via Pandas to_csv
Export filtered CSV	Write currently selected records	

The sample computations were reproduced programmatically from the submitted source file. GUI controls, external-file handling, and rendering are described based on source inspection; they should also be exercised manually on the computer used for presentation.

8. Limitations and Future Scope

Limitations: (i) the included observations are synthetic, (ii) study hours are self-reported, (iii) the dashboard is descriptive rather than predictive, (iv) the statistical thresholds are exploratory, and (v) valid student identifiers are anonymised but the application has no user accounts or access controls. Additionally, grades use a fixed threshold scheme that may not match every institution.

Possible extensions include an Add/Edit/Delete Student form, an integrated local database, assignment-specific grade thresholds, time-series tracking across semesters, downloadable summaries, handling of missing values under configurable policies, and richer analysis of possible confounding variables. Any real student dataset should be collected with appropriate consent and managed with privacy safeguards.

9. Conclusion

The project demonstrates a complete workflow for inspecting student study hours and assessment scores: CSV handling, record validation, object-oriented design, NumPy statistics, Pandas course aggregation, Matplotlib charts and a Tkinter dashboard. On the included synthetic data, a positive association persists after excluding IQR-flagged outliers, but the strength of the Pearson correlation changes substantially. The central analytical conclusion is therefore cautious: the dashboard can reveal and communicate patterns, but it cannot determine that study hours cause assessment results.

10. References

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11. Appendix: CSV Format and Execution

Requiredinput CSV headers (exactspelling): student_id,course,study_hours, score.

student_id	course	study_hours	score
S001	Mathematics	12	78
S002	Physics	8	62
S003	Electronics	15	85

To runin IDLE, open student_dashboard_idle.py and selectRun → Run Module (F5).Use Open CSV to importa file. Install required third-partypackages with: py-m pip install numpy pandasmatplotlib. The program uses a separate window for thegraphical interface.For the combined original+dashboard script, the initial Matplotlib plt.show() can delaythe interface untilearlier plot windows areclosed.